LRDTN: Spectral–Spatial Convolutional Fusion Long-Range Dependence Transformer Network for Hyperspectral Image Classification

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***Abstract*— Recently, deep learning has achieved remarkable breakthroughs in hyperspectral image (HSI) classification tasks, particularly with methods based on convolutional neural networks (CNNs) and transformers. However, these methods have several limitations: 1) the limited receptive field inherent in the convolutional layer greatly hampers capturing feature contextual information on a large scale and 2) transformers cannot establish strong local relationships, making it challenging to characterize complex dependencies between distant pixels and different bands in HSIs. Moreover, as the network complexity increases, so does the number of network parameters. We propose a novel network called the spectral–spatial convolutional fusion long-range dependence transformer network (LRDTN) for HSI classification to address these challenges. LRDTN comprises three key components: dynamic-dependent convolutional (DDC) module, the multiscale enhanced fusion (MsEF) module, and the local–global perception transformer (LGPT). Specifically, the DDC dynamically models local features, while the MsEF integrates information from different scales to capture contextual relationships in HSI features effectively. Additionally, the ability to mine and utilize HSI local–global features and complex long-range dependencies is enhanced by the proposed transformer variant, LGPT. Ultimately, through the ingeniously designed structure of the LRDTN, the model effectively maintains its performance while reducing the number of network parameters. Extensive experiments conducted on four typical HSI datasets, including urban areas, agricultural areas, and swamps, demonstrate the superiority of LRDTN over other state-of-the-art networks. The code is available at https://github.com/ybyangjing/LRDTN.**

***Index Terms*— Hyperspectral image (HSI) classification, local–global feature extraction, long-range dependence, spectral–spatial, transformer.**

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I. Introduction

HYPERSPECTRAL image (HSI) technology has garnered significant attention due to its ability to capture complex spectral–spatial dimensional information with high fidelity. It enables precise classification and identification of surface objects by leveraging the rich and continuous spectral information and spatial structural features contained in HSI [5]. Due to the extremely high spectral resolution of hyperspectral data, which enables the capture of subtle differences in the reflectance characteristics of surface objects across different spectral bands, combined with multiscale spatial texture patterns [27], scientists and engineers can reveal and distinguish various complex surface materials in numerous application scenarios [9], [31], [33], [34].

However, many pixels in hyperspectral data are affected by material mixing, which often leads to a decline in HSI classification performance [10]. Zou et al. [58] pointed out that classification methods relying solely on spectral information in HSI analysis often fail to achieve optimal classification results due to the spectral variability of the original data. To address this limitation, many classification techniques have recently emerged by fusing spectral–spatial features [1], [7], [19], [21], [37]. Nevertheless, these spectral–spatial classification strategies still have some limitations in feature extraction: the inherent limitation of the local receptive field in convolutional neural network (CNN) architectures significantly restricts the effective modeling of contextual relationships of extensive spatial feature. This limitation is particularly pronounced when dealing with surface objects that have complex neighborhood structures and spectral overlap, which can lead to constrained classification performance. For example, in Fig. 1(a), the general CNN methods fail to fully utilize the spatial contextual information of the image and lack consideration of global information. As a result, surface objects with spectral overlap yet different spatial distributions (such as corn notill, soybean notill, and soybean mintill) may be confused and misclassified.

Recently, with the popularization of the vision transformer (ViT) [13] and its many variants [44], [46] in image processing, transformer-based network models have come close to, or even surpassed, CNN-based networks in many tasks [2]. Although the transformer architecture has shown excellent performance in modeling global dependencies, it cannot handle local details. Traditional transformers cannot achieve strict local spatial constraints and spectral feature constraints

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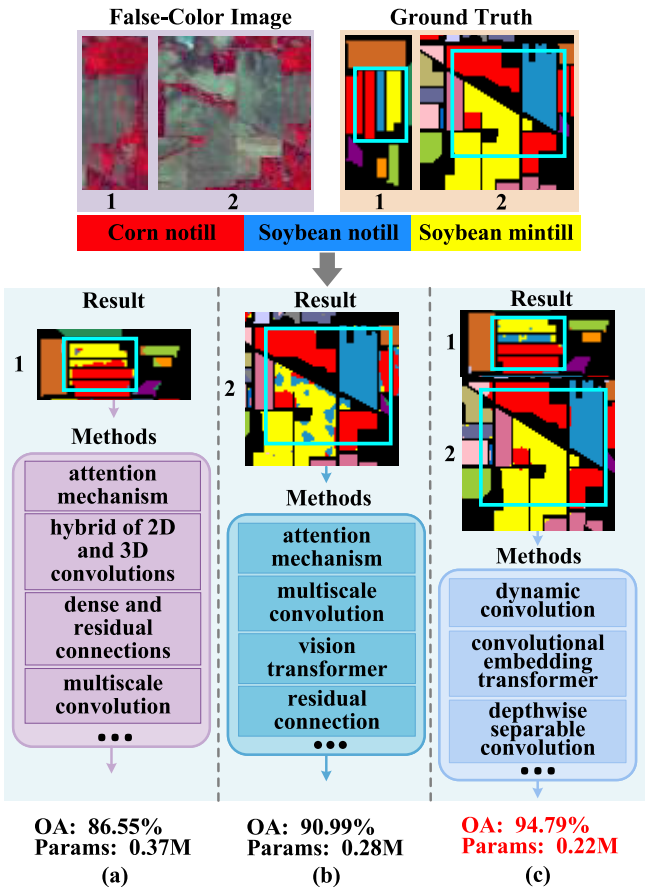


Fig. 1. Three main HSI classification methods. (a) CNN-based: fails to fully utilize the spatial context information of the image and lacks consideration of global information. (b) ViT-based: lacks an effective mechanism to synergize local and global features, and the complex structure leads to an increase in model parameters. (c) LRDTN: addresses the limitations of (a) and (b), improving precision while reducing model parameters.

on the input image data simultaneously, especially in the application of HSIs, which makes it difficult for the model to fully explore and utilize the global contextual features inherent in HSIs and their complex long-range correlations. For example, in Fig. 1(b), the model may not be able to distinguish adjacent surface objects accurately with serious spectral similarity due to the lack of an effective local and global feature synergy mechanism. In addition, introducing a transformer usually significantly increases the number of model parameters, which significantly increases the demand for computational resources. Therefore, designing a novel HSI classification framework that can fully utilize the global context and consider local details with high computational efficiency is particularly important. Therefore, a new method called long-range dependence transformer network (LRDTN) is proposed, in Fig. 1(c), which can accurately classify different surface objects with similar spectra and subtle spectral differences within the same surface object. This method achieves optimal classification performance with a relatively low number of parameters and addresses the aforementioned issues effectively.

Specifically, a dynamic-dependent convolutional (DDC) module is first proposed to effectively capture and dynamically model the spatial dependencies between different features. Second, a multiscale enhanced fusion (MsEF) module is designed to refine and fuse the rich local features extracted by the DDC module in a multilevel and omnidirectional way to enhance the overall expressive capability. Afterward, to better utilize the global contextual features embedded in HSIs and their complex long-range correlations, an encoder called a local–global perception transformer (LGPT) is proposed. Finally, we perform extensive experimental validation on four representative HSI datasets: the Salinas Valley (SV), Houston 2013 (H2013), Pavia University (PU), and Kennedy Space Center (KSC) datasets. The main contributions of the methodology in this article include the following.

1) The DDC and MsEF modules are innovatively designed to utilize the extensive spatial contextual information in HSIs effectively. The DDC module dynamically configures the convolutional kernel structure and weight allocation strategy to dynamically respond to and aggregate relevant features in spatial context changes. Building upon this, the MsEF module further extends the capability by integrating feature representations from different spatial scales, ensuring comprehensive exploration and utilization of the intrinsic spatial correlations of the image in a multiscale dimension.
2) Convolutional operations and spectral feature depth are incorporated into the multihead self-attention (MHSA) mechanism and design the multihead spectral–spatial depth (MH2SD) attention module to allow the model to integrate local features under the global view. A feature compression enhancement (FCE) module is designed to replace the traditional multilayer perceptron (MLP) module, reducing the number of network parameters while maintaining and improving model performance.
3) A new variant of the transformer architecture, the LGPT, is proposed to more deeply mine and utilize the global contextual features and complex long-range dependencies in HSIs. Based on this, the LRDTN method is constructed by integrating multiple key technology modules, including LGPT, achieving dual optimization of classification performance and parameter efficiency in HSI classification tasks.
4) Extensive experiments are conducted on the SV, H2013, PU, and KSC datasets. The experiments demonstrate that our LRDTN model achieves stable and superior classification performance with relatively small model parameter scales compared to the current state-of-the-art CNN and transformer-based methods.

II. Related Work

A. Feature Extraction Method for Spatial Context in HSI Classification

Scholars have adopted a series of innovative strategies to fully consider a wide range of contextual information in the HSI feature extraction process. For example, some studies have effectively integrated spectral and spatial information flows by constructing multilevel network structures [11], [16], [48]. Moreover, some studies have also proposed the design of multiscale feature representations [12], [38], [42], [49] to reveal the rich intrinsic structure of HSIs at different resolution

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levels. In addition, attention mechanisms are widely used [15], [18], [22], [32], [45], [50] to dynamically focus on key regions or important features to optimize further the model's ability to learn effective features. In addition, in deep learning, some studies have combined techniques such as residual and dense connectivity to construct more in-depth and extensive cellular neural network architectures for HSI classification [6], [20], [39], [54] to improve the recognition ability and generalization performance of the model for complex surface objects.

We have noticed that although general CNN-based methods consider spectral–spatial contextual information when extracting features, their inherent limited receptive fields tend to extract features within local windows. This limitation hampers their ability to capture contextual information over a broader range in hyperspectral imaging. To address this, we designed the DDC and MsEF modules to effectively utilize contextual information in HSI.

B. Feature Extraction Method Based on Transformer for HSI Classification

ViTs [13] have demonstrated powerful global modeling capabilities in processing image sequence data and achieved significant performance gains in computer vision tasks. Several research efforts have begun to explore the application of the transformer architecture to HSI classification tasks to overcome the limitations of CNN-based approaches in capturing global contextual features and long-range dependencies in HSIs. For example, Liao et al. [29] constructed spatial and spectral transformer modules to capture the spatial–spectral long-range dependencies of an image and proposed a spectral–spatial fusion transformer network (S2FTNet) for HSI classification. Hong et al. [23] considered HSI classification from the perspective of sequences, proposed a new backbone network called SpectralFormer that learns spectral knowledge from grouped neighboring spectral bands, and proposed a module called cross-layer adaptive fusion (CAF). Although the global self-attention (SA) mechanism enables ViT to capture and model long-range dependencies in images efficiently, transformers are slightly less effective than CNNs in directly constructing local spatial structures and details. Therefore, researchers have proposed a variety of novel HSI classification frameworks that combine the advantages of ViT and CNNs by fusing the properties of both to achieve more comprehensive feature extraction [24], [47], [51], [53], [56].

We have observed that the aforementioned methods merely represent a simple combination of CNNs and transformers in form, without fully leveraging the advantages of both for deep integration. This may lead to incoherent information flow and redundant feature representation. Additionally, the introduction of ViT significantly increases the model parameters, thereby raising computational costs. Based on this, we propose a new variant of the transformer architecture—LGPT—aimed at establishing the ability to process local relationships, uncover complex dependencies between different bands, and reduce model complexity.

III. Methods

In this section, we have provided a detailed description of the proposed LRDTN. Specifically, batch normalization (BN) and gaussian error linear unit (GeLU) activation are added to our network framework after each convolution. For the sake of brevity, it is not specified in the following description.

A. Overview of the LRDTN

We propose a spectral–spatial convolutional fusion LRDTN with a dual-branch architecture for HSI classification to effectively capture the contextual relationships of features in HSIs and enhance the ability of the model to extract and utilize local–global features and complex long-range dependencies. The model structure is illustrated in Fig. 2. The LRDTN consists of four stages, where the input 3-D HSI data $\mathcal{I}' \in \mathcal{R}^{\{C\times H\times W\}}$ are first segmented into patches $\mathcal{I} \in \mathcal{R}^{\{C\times h\times w\}}$ to utilize better spectral and spatial information, where C and $H \times W$ denote the number of spectra and spatial size, respectively, and $h \times w$ is the patch size. The second stage takes the patch $\mathcal{I}$ as input to the dynamic context feature extraction block (Branch 1) for processing to dynamically extract the multiscale contextual relation feature O_{up}

$$O_{\text{up}} = P\left(\text{Conv}^{C\to 128}(\text{MsEF}(\text{DDC}(\mathcal{I}) + \mathcal{I}) + \mathcal{I})\right) \quad (1)$$

where $P(\cdot)$ denotes the adaptive average pooling operation. $\text{Conv}^{C\to 128}$ is a convolution operation where the convolution kernel is 1×1 and the output channel becomes 128.

The third stage also takes patch $\mathcal{I}$ as input to the local–global feature extraction block (Branch 2), which is processed to extract the local–global feature O_{down} with long-range dependence. In Branch 2, to ensure that the converter can process the image data efficiently, the input patch block is flattened and projected to the sequence data with specific channel dimensions, which can be represented by the following equation:

$$T_{\text{PF}} = \text{Projection}(\text{Flatten}(\mathcal{I})) \quad (2)$$

where T_{PF} is the sequence of data after flattening and projection. Afterward, the position information P_o is encoded into the sequence feature T_{PF} to obtain the input T_{tran} of the LGPT encoder

$$T_{\text{tran}} = T_{\text{PF}} + P_o. \quad (3)$$

The unsqueeze() function is used to perform feature fusion with Branch 1 and change the shape of the output tensor of the LGPT encoder

$$O_{\text{down}} = \text{unsqueese}(\text{LGPT}(T_{\text{tran}})). \quad (4)$$

In the fourth stage, the output features of branches 1 and 2 are cascaded and injected into a 1×1 convolution for feature generalization, improving the feature representation of the model. Finally, the summarized features are fed into an output layer consisting of a fully connected layer and a softmax function. This is expressed using the following equation:

$$O_{\text{out}} = \text{Softmax}(f(\rho([O_{\text{up}}; O_{\text{down}}]))) \quad (5)$$

where [;] denotes the operation of connecting multiple features, $\rho(\cdot)$ is the feature induction performed, $f(\cdot)$ denotes the feature learning through the fully connected layer, and O_{out} is the categorization probability of different categories.

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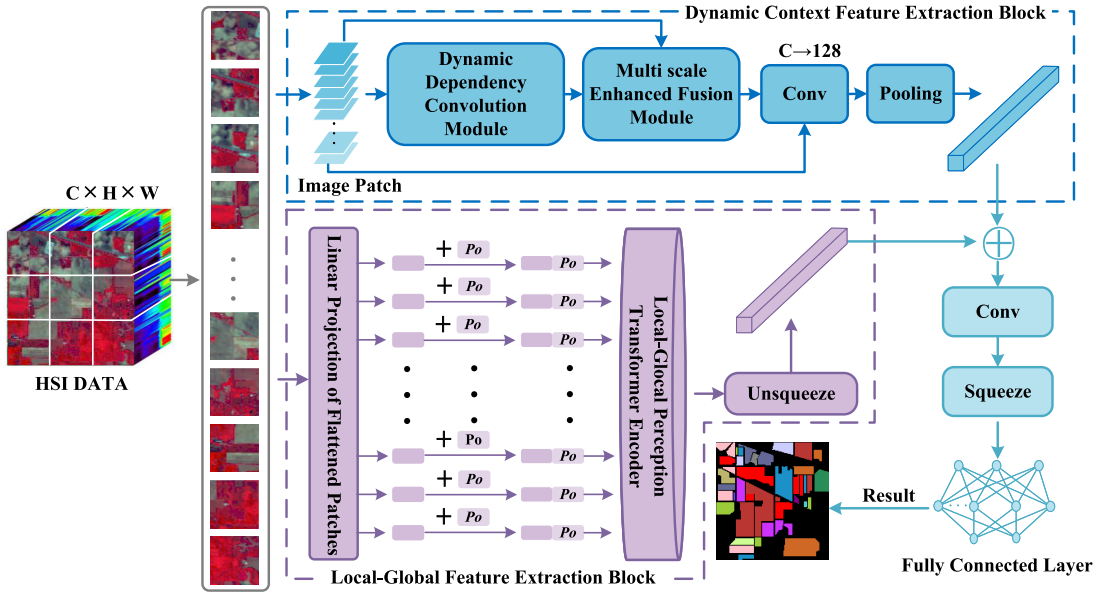


Fig. 2. Proposed LRDTN classification network consists of four stages. Stage 1, the HSI data are segmented into patches; stage 2, extract multiscale contextual relationship features from HSI; stage 3, extract local–global information with long-range dependencies from HSI; and stage 4, the features obtained from branches 1 and 2 are fused.

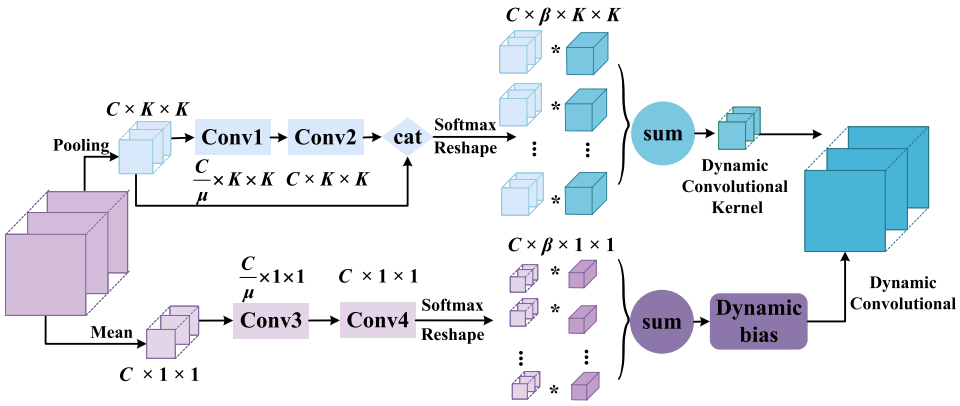


Fig. 3. Detailed structure of the DDC module.

B. Dynamic Context Feature Extraction Block

The structure of the dynamic context feature extraction block is shown in Fig. 2, which comprises the DDC and the MsEF module, through which we extract more comprehensive, rich, and diversified feature representations and improve the contextual relationship modeling between features. We describe these two modules in detail.

1) Dynamic-Dependent Convolution: We propose a novel DDC better to model feature context relationships, as shown in Fig. 3. First, let the input feature map be $\mathcal{I} \in \mathcal{R}^{\{C\times h\times w\}}$, and then, use adaptive average pooling to aggregate the spatial context to obtain the feature map $X_{\text{pool}} \in \mathcal{R}^{\{C\times K\times K\}}$. Then, important features are extracted to reduce computational costs, followed by feature compression to reduce dimensions and feature reconstruction and restore dimensions, resulting in an attention map $T' \in \mathcal{R}^{\{C\times K\times K\}}$. The equation is as follows:

$$T' = \text{Conv}_2^{\frac{C}{\mu}\to C}\left(\text{Conv}_1^{C\to\frac{C}{\mu}}\left(\text{Pool}^{C\times h\times w\to C\times K\times K}(\mathcal{I})\right)\right) \quad (6)$$

where Pool$(\cdot)$ is the adaptive average pooling layer, and $\text{Conv}_1(\cdot)$ and $\text{Conv}_2(\cdot)$ denote the feature compression and feature reconstruction using 1×1 convolution, respectively. μ is the scaling factor, based on relevant studies [18] and the analysis of systematic parameter optimization experimental results over the range of μ values {1, 2, 4, 8, 16}, ultimately determining that $\mu = 2$.

Next, to enhance the representation and comprehension of the input data in the model, T' is spliced with $\mathcal{I}$ and reshaped to $T' \in \mathcal{R}^{\{C\times\beta\times K\times K\}}$. Then, the softmax function is used to obtain the attention weight T, as shown in the following equation:

$$T = \text{Softmax}\left(\text{Re}^{\{C\times K\times K\}\to\{C\times\beta\times K\times K\}}\left(\text{cat}\left(T', \mathcal{I}\right)\right)\right) \quad (7)$$

where cat$(\cdot)$ denotes the splicing of T' with $\mathcal{I}$ according to the channel dimension. β is the number of attention groupings. Re$(\cdot)$ represents the reshape. Afterward, T is multiplied by a set of learnable parameters $\mathcal{M} \in \mathcal{R}^{\{C\times\beta\times K\times K\}}$, and the outputs are summed over the β-dimension to obtain the dynamic

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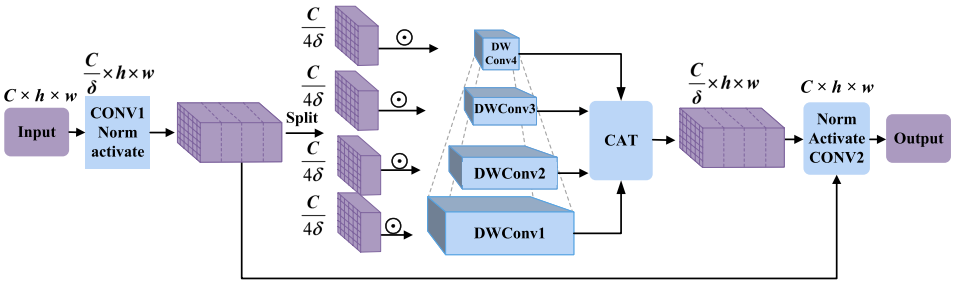


Fig. 4. Detailed structure of the MsEF module.

convolution (DyConv) kernel $\mathcal{K}$ that depends on the input data, $\mathcal{K} \in \mathcal{R}^{\{C\times K\times K\}}$

$$\mathcal{K} = \text{sum}_\beta(T \circledast \mathcal{M}) \tag{8}$$

where $\text{sum}_\beta(\cdot)$ denotes the summation in the β-dimension and $\circledast$ denotes the multiplication. In addition, we perform a softmax function on the input data $\mathcal{I} \in \mathcal{R}^{\{C\times h\times w\}}$ after feature expansion and then multiply it with a set of learnable parameters $\mathbb{R} \in \mathcal{R}^{\{C\times\beta\}}$ to obtain a dynamic bias $\mathcal{B}$ that depends on the input data

$$\mathcal{B} = \text{Softmax}\left(\text{Conv}_4^{\frac{C}{\mu}\to C}\left(\text{Conv}_3{}^{C-\frac{C}{\mu}}(\text{mean}(\gamma))\right)\right) \circledast \mathbb{R} \tag{9}$$

where $\text{Conv}_3(\cdot)$ and $\text{Conv}_4(\cdot)$ denote the feature compression and feature expansion using 1×1 convolution, respectively. Mean$(\cdot)$ represents calculating the average value based on a certain dimension. μ is the scaling factor, and the selection of μ can be referenced in (6); here, $\mu = 2$.

Finally, the input $\mathcal{I} \in \mathcal{R}^{\{C\times H\times W\}}$ is convolved with the dynamic-dependent convolution to obtain the module output

$$Y_D = (\mathcal{I} \times \mathcal{K}) + \mathcal{B}. \tag{10}$$

Compared to the static nature of traditional convolution, the DDC module dynamically models the dependencies between features by adaptively adjusting the shape and weight of the convolution kernel to perform local feature aggregation in an input-dependent manner. This ability to model dynamic dependencies enables the DDC module to better capture the contextual relationships in the input data and achieve more dynamic local feature modeling, improving the performance and generalizability of the model in complex visual tasks.

2) Multiscale Enhanced Fusion (MsEF): As shown in Fig. 4, we propose a MsEF module, which aims to refine and fuse the rich local features extracted by the DDC module in a multilevel way to enhance the overall expression capability. In HSI classification tasks, the classification of pixels often relies on the contextual information of their surrounding regions. MsEF performs convolutional operations at different scales and combines features at different scales through residual joining to utilize better the contextual information. This enables the model to more accurately understand the relationship between pixels and their surroundings in the image and obtain rich contextual information from them. We use separable convolution instead of traditional convolution to better reduce the computational complexity of the model in terms of parameters. Assume the input to the model is HSI data $U \in R^{C\times h\times w}$.

First, to reduce the computational and storage overhead, we perform a channel reduction operation on the input features

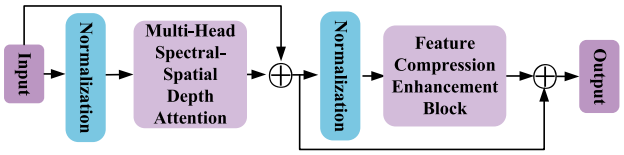


Fig. 5. Detailed structure of the designed LGPT encoder.

to obtain the output features $U' \in \mathbb{R}^{((C/\delta)\times h\times w)}$, where δ is the scaling factor; here, $\delta = 6$. Afterward, the scaling factor is segmented into multiple subfeature maps according to the channel dimensions, and multiscale convolution is performed separately. The multiscale convolution can capture the feature information at different scales, which helps to extract the local and global features in the image, making the model multiscale aware. We define a set of multiscale convolution kernels $D = \{D_1, D_2, D_3, \ldots, D_k\}$, $k = 4$, and the size of the kernel is $C_k \times C_k$, $C_k = \{1, 3, 5, 7\}$ (parameter selection analysis in Section IV-E2). The subfeature maps are convolved separately with a multiscale convolutional kernel to extract the channel-related features T_k with neighboring distances C_k given a kernel size of $C_k \times C_k$. The equation is as follows:

$$U' = \delta(U) \tag{11}$$

$$T_k = \text{Split}_C{}^4(U') \odot D_k \tag{12}$$

where $\odot$ is the convolution operation, D_k is the multiscale convolution kernel, and $\text{Split}_C{}^4$ denotes the partitioning of U' into four subfeature maps according to the channel dimension. The channel-related features T_k and adjacent distance C_k are fused; the fused features T_s are represented as follows:

$$T_s = \sum_{d=1}^{g} T_k^d \tag{13}$$

where T_k^d denotes the feature obtained after the convolution operation with the dth multiscale convolution, and we used four convolutions of different sizes, so $g = 4$.

Afterward, to integrate the feature information at different scales, enhance the feature context perception, and preserve the richness of the multiscale features, we connect the fused features T_s with U' for residual concatenation and channel recovery. The equation is as follows:

$$Y_{\text{Ms}} = \frac{1}{\delta}(T_s + U'). \tag{14}$$

C. Local–Global Perception Transformer Encoder

As shown in Fig. 5, we propose a transformer variant called LGPT to utilize better the global contextual features embedded in HSIs and their complex long-range dependencies.

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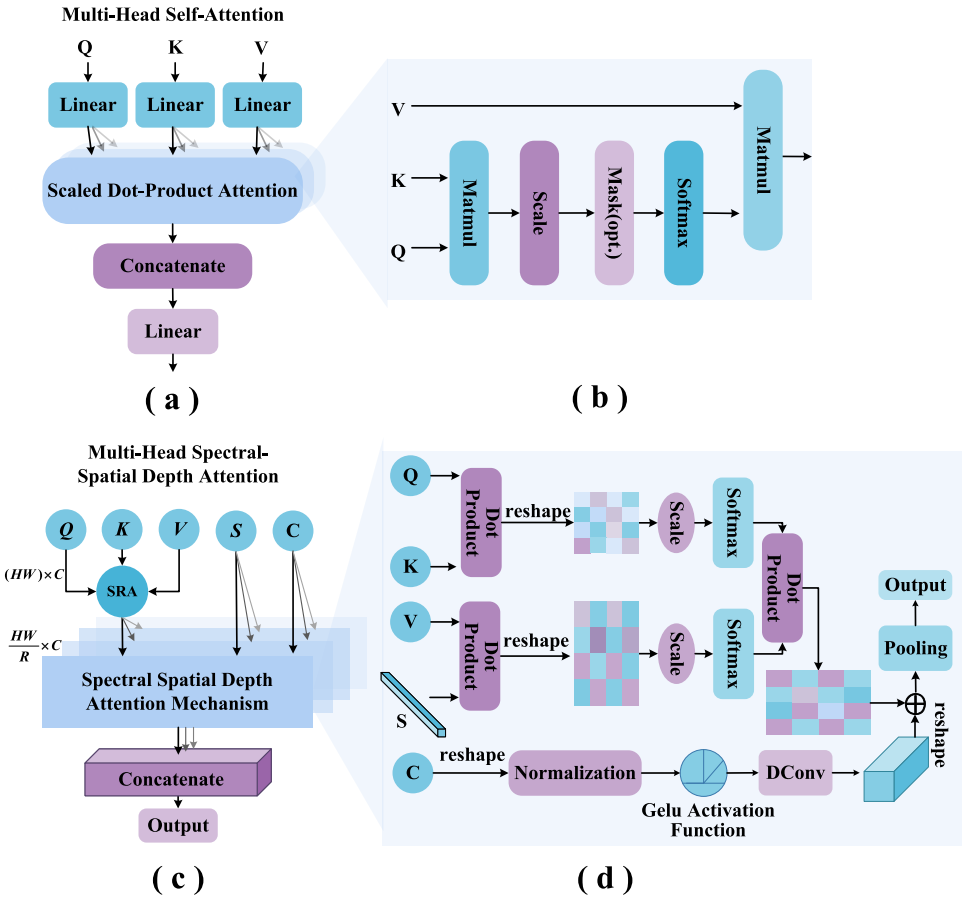


Fig. 6. Attention mechanism structure in traditional ViT and the proposed LRDTN. Traditional attention mechanisms: (a) MHSA and (b) SA. (c) Proposed attention mechanism is MH2SD attention. (d) 2SD attention mechanism.

The proposed LGPT comprises MH2SD attention and an FCE block. In the LGPT module, the input feature T_{tran} is first normalized; then, the features are input to MH2SD for feature processing to obtain contextual information and better model the local information; afterward, the output feature T_{MH2SD} of MH2SD is residually concatenated with the input T_{tran}, and the output is normalized; then, the features are transferred to the FCE block for processing to realize the deep fusion of local and global information. Finally, the output T_{FCE} is residually concatenated to obtain the output of the LGPT

$$T_{\text{MH2SD}} = \text{MH2SD}(N(T_{\text{tran}})) \tag{15}$$

$$F_{\text{lgpt}} = \text{FCE}(N(T_{\text{MH2SD}} + T_{\text{tran}})) + T_{\text{MH2SD}} + T_{\text{tran}} \tag{16}$$

where $N(\cdot)$ denotes the normalization. T_{tran} represents the input features of LGPT, and (3) provides detailed source explanations. MH2SD$(\cdot)$ and FCE$(\cdot)$ represent the feature processing operations of the MH2SD and FCE modules, respectively. T_{MH2SD} denotes the output feature of the MH2SD module. Next, we describe the MH2SD and FCE modules in detail.

1) MH2SD Attention: As shown in Fig. 6(a) and (b), a traditional MHSA [11] can be described as a mapping of a query Q and a set of key K-value V pairs to an output, where the SA layer can be represented as

$$Y_y = \text{SA}(Q, K, V) = \text{Softmax}\left(\frac{QK^{\text{T}}}{\sqrt{d}}\right)V \tag{17}$$

where d denotes the dimensions of Q and K, and Y_y is the output of SA.

However, pure ViT cannot utilize the local feature information of HSIs as effectively as CNNs can. Therefore, we compensate for the lack of local modeling by introducing convolutional operations in ViT. The module can better capture local features in the image by introducing depth-separable convolution, focusing on the details and local structure of the image, which helps to improve the perception of complex textures and local features in HSIs. Moreover, to make the attention operation more spectrally aware, we introduce spectral information in ViT to fully utilize the spectral response of different bands in HSIs. The structure of the proposed MH2SD with spectral–spatial information is shown in Fig. 6(c). We assume that the input feature is T_{in}. First, T_{in} is converted into a query feature map Q, a key feature map K, and a value feature map V by convolutional mapping. Afterward, we introduced the spatial reduction attention (SRA) module [43], which receives Q, K, and V as inputs and reduces the spatial scale through the SRA module to realize feature compression by removing redundant or unimportant information and reducing the number of parameters

$$Q, K, V = \text{SRA}(\text{Conv}_{1\times1}(T_{\text{in}})) \tag{18}$$

where $\text{Conv}_{1\times1}(\cdot)$ denotes a convolution operation with a convolution kernel of 1×1, and SRA$(\cdot)$ is an

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operation used to reduce the spatial dimension of the input sequence.

We also supplement the original MHSA with depth-separable convolution C to capture local features. Meanwhile, the spatial features of each channel are integrated channel-by-channel by a global pooling operation to generate a global feature to obtain the spectral feature S of the feature T_{in}, $S \in R^C$

$$S_C = \frac{1}{h \times w} \sum_{i=0}^{h-1} \sum_{j=0}^{w-1} P(i, j) \tag{19}$$

where S_c is the global eigenvalue of channel c, and $P(i, j)$ is the value of feature T_{in} at position (i, j) for channel c.

By improving the traditional MHSA, the proposed spectral–spatial depth (2SD) attention mechanism is shown in Fig. 6(d) and can be defined as

$$T = \left(\xi\left(\frac{QK^{\text{T}}}{\sqrt{d}}\right)\xi\left(\frac{SV}{\sqrt{d}}\right)\right) + D_{\text{conv}}\left(N(G(C))\right) \tag{20}$$

where T is the 2SD output, $G(\cdot)$ is the GeLU activation function, $N(\cdot)$ is the normalization layer, $D_{\text{conv}}(\cdot)$ is the depth-separable convolution operation, and $\xi(\cdot)$ is the Softmax function. Afterward, a pooling layer maps T to the same dimension as the input features. The 2SD can capture the features of HSIs more accurately and improve the ability to discriminate between different classes by fusing spectral information with spatial information.

In the proposed MH2SD, different heads work independently in parallel mode to learn different attentions, as shown in Fig. 6(c). The attentions are connected to obtain the output T_{MH2SD}.The proposed MH2SD with h heads can be represented as

$$T_{\text{MH2SD}} = \text{Concat}\left(T^{(1)}, T^{(2)}, \ldots, T^{(h)}\right) \tag{21}$$

where h denotes the number of heads of attention. We conducted a detailed experimental analysis of the selection of h in Section IV-D1.

In summary, the LGPT integrates multihead attention mechanisms, depthwise separable convolution (DwsConv), and spectral features to overcome the inherent limitations of transformers in handling local spatial features, enhancing the understanding of the intrinsic long-range dependencies in HSIs.

2) Feature Compression Enhancement: After applying the multihead attention mechanism, traditional methods use an MLP for feature extraction. However, MLPs lack local perception. An MLP is a fully connected feedforward neural network that globally maps input features without explicit local awareness, and it has many parameters. We replace the original MLP in the MHSA with an FCE block to achieve deep fusion between the CNN and ViT while reducing the number of network parameters, as depicted in Fig. 7. Given the input features F, we first reshape them and then utilize deep convolution to enhance local relationships

$$F_D = D_{\text{conv}}(F). \tag{22}$$

We use 1×1 convolution to compress and expand the channel of feature F_D and reuse features through residual connections

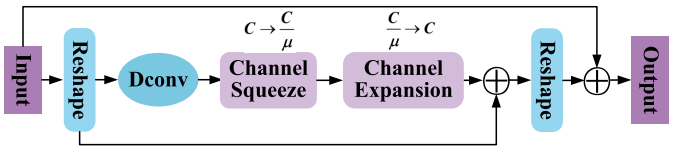


Fig. 7. Detailed structure of the FCE module.

to reduce computational costs. The output F_{fce} of FCE is represented as follows:

$$F_{\text{fce}} = \text{FCE}(F) = C_{\text{ce}}{}^{\frac{C}{\mu} \to C}\left(C_{\text{cs}}{}^{C \to \frac{C}{\mu}}(F_D)\right) + F \tag{23}$$

where μ is the scaling factor, and the selection of μ can be referenced in (6); here, $\mu = 6$.

D. Training

The training process of LRDTN is shown in Algorithm 1. First, the input data $\mathcal{I}$ are passed to the DDC module, where the contextual relationships of local features Y_D are dynamically modeled through (6)–(10). Next, Y_D is input into the MsEF module to obtain multilevel local features Y_{Ms} using (11)–(14). Then, multiscale contextual relationship features O_{up} are generated through (1). Meanwhile, the input data $\mathcal{I}$ are passed to the LGPT module, which captures the global contextual features and their complex long-range dependence relationships inherent in HSI through (15) and (16). Finally, local–global features O_{down} with long-range dependencies are generated using (4). The final step involves aggregating the features O_{up} and O_{down}.

Algorithm 1 Training Process

Input:
3D HSI data $\mathcal{I} \in \mathcal{R}^{\{C \times h \times w\}}$; number of epochs T;
learning rate = 0.0005.
Output:
the prediction vector of target data O_{out}, compute OA, AA, and Kappa.

Begin
for *each* T **do**
  DDC dynamically models the contextual relationship Y_D of local features through Eqs. (6)-(10);
  MsEF obtain multilevel local features Y_{Ms} through Eqs. (11)-(14);
  Generate multiscale contextual relationship features O_{up} through Eq. (1);
  LGPT captures global context features and complex long-term dependencies through Eqs. (15)-(16);
  Generate local-global features O_{down} with long-range dependencies through Eq. (4);
  Feature Induction: $Z_{out} = \rho([O_{up}; O_{down}])$;
end
$O_{out} = Classifier(Z_{out})$.
Compute OA, AA, and Kappa.
End

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR SV AND H2013 DATASETS

No.	SV				H2013			
	Class name	Total	Train	Test	Class name	Total	Train	Test
1	Broccoli green weeds 1	2009	30	1979	Grass healthy	1251	40	1211
2	Broccoli green weeds 2	3726	30	3696	Grass stressed	1254	40	1241
3	Fallow	1976	30	1946	Grass synthetic	697	40	657
4	Fallow rough plow	1394	30	1364	Tree	1244	40	1204
5	Fallow smooth	2678	30	2648	Soil	1242	40	1202
6	Stubble	3959	30	3929	Water	325	40	285
7	Celery	3579	30	3549	Residential	1268	40	1228
8	Grapes untrained	11271	30	11241	Commercial	1244	40	1204
9	Soil vinyard develop	6203	30	6173	Road	1252	40	1212
10	Corn senesced green weeds	3278	30	3248	Highway	1227	40	1187
11	Lettuce romaine 4wk	1068	30	1038	Railway	1235	40	11195
12	Lettuce romaine 5wk	1927	30	1897	Parking lot 1	1233	40	1193
13	Lettuce romaine 6wk	916	30	886	Parking lot 2	469	40	429
14	Lettuce romaine 7wk	1070	30	1040	Tennis court	428	40	388
15	Vinyard untrained	7268	30	7238	Running track	661	40	620
16	Vinyard vertical trellis	1807	30	1777	-	-	-	-
Total	-	54129	480	53649	-	15029	6000	14429

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR PU AND KSC DATASETS

No.	PU				KSC			
	Class name	Total	Train	Test	Class name	Total	Train	Test
1	Asphalt	6631	30	6601	Scrub	761	20	741
2	Meadows	18649	30	18619	Willow swamp	243	20	223
3	Gravel	2099	30	2069	CP hammock	256	20	236
4	Trees rough plow	3064	30	3034	CP/Oak	252	20	232
5	Metal sheets	1345	30	1315	Slash pine	161	40	141
6	Bare soil	5029	30	4999	Oak/Broadleaf	229	20	209
7	Bitumen	1330	30	1300	Hardwood swamp	105	20	85
8	Bricks	3682	30	3652	Graminoid marsh	431	20	411
9	Shadows	947	30	917	Spartina marsh	520	20	500
10	-	-	-	-	Catiail marsh	404	20	384
11	-	-	-	-	Salt marsh	419	20	399
12	-	-	-	-	Mud flats	503	20	483
13	-	-	-	-	water	927	20	907
Total	-	42776	270	42506	-	5031	260	4771

IV. EXPERIMENTS AND ANALYSIS

In this section, numerous experiments are conducted on the model to demonstrate the effectiveness of the proposed method.

A. Datasets Description

Four widely used HSI benchmark datasets are selected for comparative experiments to evaluate the performance of the algorithm proposed in this article: SV [28], H2013 [52], PU [40], and KSC [59]. The details of the datasets are shown in Tables I and II.

1) SV Dataset: The SV dataset is a classic application case of HSI technology in remote sensing image processing for agriculture. It was collected by the National Aeronautics and Space Administration (NASA) in 1993 in SV, CA, USA. It consists of 204 bands of size 512 × 217. The dataset contains hyperspectral remote sensing images of agricultural

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growing areas in the SV region and 16 ground-truth classes, such as corn, beans, oilseed rape, maize, and land.

2) H2013 Dataset: The H2013 dataset is a classic application case of HSI technology in remote sensing image processing of urban areas. It was collected by NASA in 2013 in the urban area of Houston, TX, USA. It includes 144 bands of size 349 × 1905 and comprises 15 ground-truth classes such as residential areas, parks, roads, and buildings.

3) PU Dataset: The PU dataset is a classic application case of HSI in remote sensing image processing of urban areas. This dataset was acquired by the German Aerospace Center (DLR) using an airborne reflectance optical spectral imager (ROSIS-03) to image the University of Pavia campus in Italy in 2003. It has 103 bands of size 610 × 340 and includes nine ground-truth classes, such as asphalt, meadows, and gravel.

4) KSC Dataset: The KSC dataset is a use case of HSI technology in remote sensing image processing of swamp areas. It was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), FL, USA, in 1996, and consists of 224 bands of size 512 × 614. We used 176 of these bands after removing the bands with water absorption and a low signal-to-noise ratio. There are 13 ground-truth classes, including the scrub, spartina marsh, and water classes.

B. Experimental Settings

1) Evaluation Setup: We use several broad metrics to evaluate the performance of the LRDTN, including overall accuracy (OA), average accuracy (AA), kappa coefficient (Kappa), training time, and testing time.

2) Environmental Setup: In our experiments, we repeated each experiment five times and reported the average result and standard deviation of each metric to avoid the effect of random operation. The LRDTN network proposed in this article and the other comparison networks are implemented on the PyTorch platform. All networks are run on an i7-11700k CPU, an RTX3080Ti GPU, 32 GB of RAM, and a 64-bit Windows 11 operating system software environment.

3) Parameter Setup: We ran the code of seven classification methods published online and attempted to achieve the highest classification accuracy in the four hyperspectral datasets with the same hyperparameter settings for all the methods. Adam optimizer [25] is used to optimize the network parameters with a batch size of 100. The number of epochs is set to 200 based on a series of studies [4], [12], [14], [26]. The learning rate and patch size are set to 5e−4 and 11 × 11, respectively. The number of layers in the transformer encoder and the number of heads in the MHSA are set to 1 and 4, respectively. In Section IV-D, the experimental analyses are conducted on some of the hyperparameters to validate the rationale behind their selection.

4) Comparison Methods: Traditional CNN methods are limited in their ability to capture contextual information at a large scale due to the inherent finite receptive field of convolutional layers. And transformer-based methods lack the capability to establish strong local relationships, making it challenging to describe the complex dependencies within HSI data. Therefore, to validate the capability of the proposed LRDTN in capturing HSI feature context and leveraging its complex long-range dependencies, four state-of-the-art CNN-based methods and three recent transformer-based methods are selected for comparative experiments. Detailed descriptions of the comparison methods are provided as follows.

1) *SSRN [55]:* Uses 3-D convolution-based spectral and spatial residual blocks to extract spectral and spatial features.
2) *A2S2K [35]:* Learns selective 3-D convolutional kernels and uses an improved 3-D ResBlock to jointly extract spectral features for HSI classification.
3) *GMA_NET [30]:* Embedding multiple attention mechanisms in CNN to explore important information within groups and propagating differential information between groups to learn spectral–spatial features of specific groups.
4) *MTGAN [17]:* Uses a CNN-based generator and discriminator to simultaneously reconstruct and classify input data, thereby improving the classification performance of the CNN.
5) *ViT [13]:* A classic transformer network that is a pure ViT without any convolutional operations.
6) *MF [36]:* A novel transformer variant that uses spectral and spatial morphological convolution operations (combined with attention mechanisms) to improve the interaction between the structure and shape information of HSI tokens and the classification (CLS) token for HSI classification.
7) *LSFAT [41]:* A novel transformer variant that uses neighborhood aggregation-based embedding (NAE) and neighborhood attention (NAA) modules to adaptively form multiscale features and capture long-range dependencies between features.

C. Comparison With the State-of-the-Art Methods

Tables III–VI present the detailed results; the corresponding visualizations are shown in Figs. 8–11. (The bold text represents the results of the proposed method.) (In the KSC dataset, we did not perform an experimental comparison of this method due to the special design of the grouped multi-attention network (GMA-NET) method, which resulted in the inability to reproduce the results of GMA-NET under the KSC dataset.)

1) Table III and Fig. 8 Show the Classification Results and Visualized Classification Maps for the SV Dataset, Respectively: As shown in Table III, LRDTN outperforms the other compared methods in terms of OA, AA, and Kappa, especially for classes 7 (celery) and 12 (lettuce romaine 5 wk); the classification accuracy of the LRDTN method reaches 99.99%. In addition, the transformer-based method yields better classification results than does the CNN-based method. Among the transformer-based methods, ViT has the worst classification effect, which can be attributed to the fact that the MF, LSFAT, and LRDTN methods introduce convolutional operations in transformers, which allows the model to compensate for the shortcomings of transformers in localized feature extraction, thus improving the classification performance. By looking at the visualization results in Fig. 8, we find that for vineyard untrained and grapes untrained, which are the two most easily observed categories, the LRDTN

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TABLE III
CLASSIFICATION RESULTS OF THE SV DATASETS

No.	CNN-based method				ViT-based method			
	SSRN	A2S2K	GMA_NET	MTGAN	ViT	MF	LSFAT	LRDTN
1	98.45±3.24	90.35±19.13	100.00±0.00	100.00±0.00	94.45±4.27	98.47±2.14	99.28±1.55	**97.20±3.16**
2	89.83±10.20	98.51±1.00	96.19±4.67	99.68±0.28	89.42±18.84	99.52±0.35	97.78±1.74	**99.63±0.45**
3	66.36±24.09	86.82±15.96	96.38±5.12	90.17±19.78	69.02±16.58	99.64±0.48	96.29±3.74	**99.52±0.67**
4	98.55±2.23	97.46±3.89	99.72±0.38	99.77±0.34	95.50±1.55	99.00±1.29	99.81±0.11	**99.60±0.48**
5	57.79±24.14	78.56±28.40	98.19±0.98	99.37±0.31	95.15±0.91	96.36±3.19	93.25±5.15	**99.61±0.38**
6	99.73±0.53	99.23±1.10	99.80±0.08	99.99±0.01	100.00±0.00	99.99±0.01	99.99±0.01	**99.97±0.03**
7	99.68±0.19	99.62±0.33	99.79±0.15	99.97±0.03	98.59±1.09	99.86±0.10	99.81±0.12	**99.99±0.01**
8	58.54±16.52	69.81±10.97	86.21±2.56	87.16±5.68	62.59±12.70	80.03±3.87	77.04±2.55	**84.30±1.16**
9	96.86±1.29	94.13±6.10	99.92±0.09	98.11±2.49	97.56±1.05	99.33±0.78	97.50±1.37	**99.70±0.43**
10	86.98±7.19	94.56±4.70	95.97±2.77	97.07±0.82	73.06±17.99	97.70±0.83	93.94±1.76	**95.95±1.67**
11	85.40±15.94	87.84±14.53	86.28±26.33	100.00±0.00	92.97±2.58	99.23±1.67	98.83±1.21	**99.29±1.13**
12	67.00±32.29	99.80±0.39	99.97±0.07	99.20±1.16	98.99±1.98	98.68±2.06	99.72±0.56	**99.99±0.02**
13	98.49±2.52	94.06±9.98	99.09±0.21	99.68±0.59	97.11±3.76	99.91±0.13	99.93±0.15	**99.75±0.55**
14	98.17±1.19	96.85±2.55	97.29±1.66	95.62±7.36	97.85±0.88	98.64±0.68	96.89±3.17	**99.44±0.65**
15	95.79±3.33	88.47±8.80	84.74±5.40	80.52±8.93	80.86±13.75	91.46±2.86	73.66±7.05	**94.21±2.22**
16	91.14±7.14	82.95±29.74	99.22±1.64	99.24±0.48	88.68±7.57	99.01±0.72	97.73±1.09	**99.37±0.59**
OA(%)	83.66±4.44	88.01±2.54	93.92±0.55	93.72±0.51	84.53±2.85	94.01±0.40	90.15±0.36	**95.42±0.08**
AA(%)	86.80±4.46	91.19±4.19	96.17±1.50	96.60±1.22	89.49±2.03	97.30±0.21	95.09±0.23	**97.97±0.16**
Kappa(%)	81.91±4.88	86.71±2.79	93.23±0.61	93.01±0.57	82.86±3.10	93.35±0.43	89.05±0.41	**94.91±0.09**

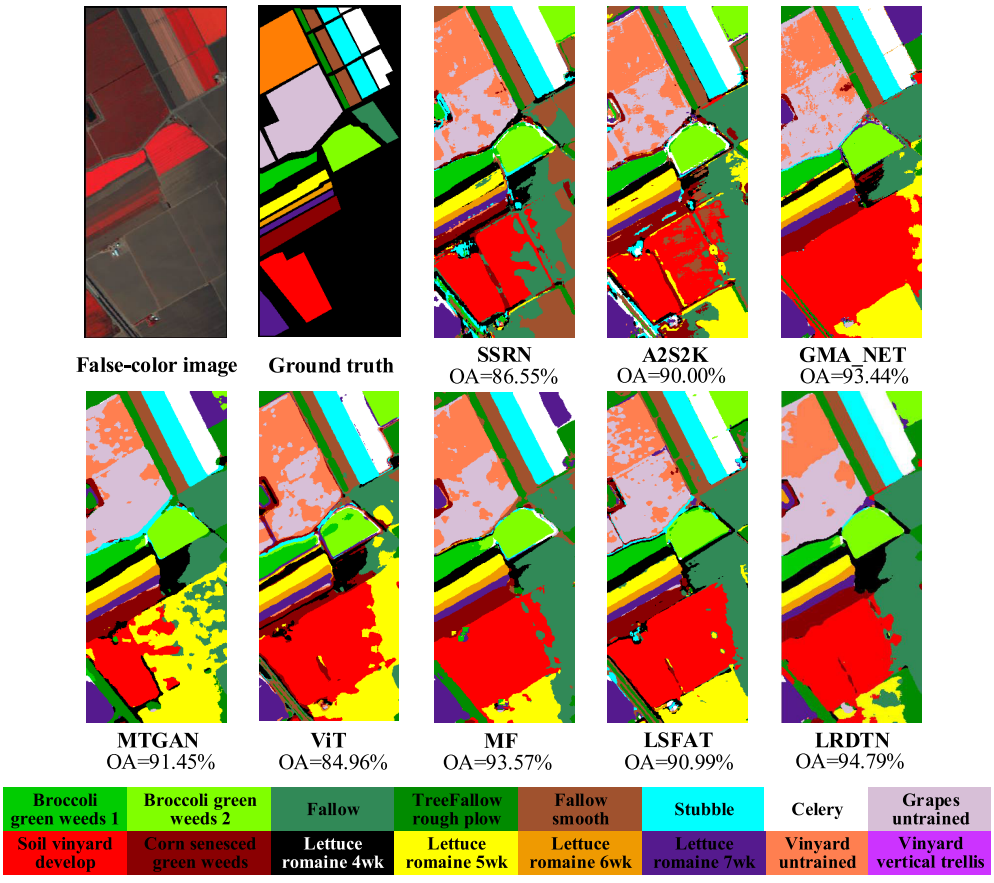


Fig. 8. Classification maps of different methods on the SV dataset.

exhibits the best visual results even though it does not obtain the highest classification accuracies because the LRDTN performs well in capturing visual features that may not directly correlate to pure classification accuracy metrics yet produce

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TABLE IV
CLASSIFICATION RESULTS OF THE H2013 DATASETS

No.	CNN-based method				ViT-based method			
	SSRN	A2S2K	GMA_NET	MTGAN	ViT	MF	LSFAT	LRDTN
1	96.96±2.61	93.43±3.41	98.70±1.56	84.66±1.11	91.03±4.74	87.15±5.67	90.88±4.83	**82.94±2.34**
2	79.59±23.88	94.58±6.47	89.74±5.65	99.82±0.12	97.89±1.57	95.34±5.46	99.01±0.70	**97.27±4.58**
3	96.13±5.31	75.31±31.49	100.00±0.00	99.97±0.07	99.57±0.20	99.48±0.84	100.00±0.00	**99.94±0.08**
4	30.32±11.72	54.07±25.68	94.35±0.00	93.69±1.10	90.95±0.95	97.19±1.42	89.67±6.43	**99.14±0.57**
5	83.64±10.42	65.72±33.87	99.17±0.81	99.90±0.18	99.50±0.38 9	9.30±1.23	99.37±0.47	**99.87±0.25**
6	83.44±21.03	68.49±26.63	99.72±0.63	86.67±0.00	91.30±5.31	99.09±1.21	97.33±1.90	**86.46±0.47**
7	51.43±14.76	41.22±14.89	90.90±3.38	87.35±2.81	81.47±2.48	81.19±8.33	82.30±4.28	**88.19±2.43**
8	70.27±9.54	69.10±11.15	83.12±4.15	85.50±2.08	74.49±3.43	83.41±8.09	79.05±1.97	**86.94±1.77**
9	25.26±8.05	30.54±8.58	91.73±3.22	92.04±2.20	85.41±2.38	79.24±11.51	89.26±2.94	**91.22±1.69**
10	61.21±22.47	39.32±17.84	69.38±1.82	74.32±7.64	73.46±5.01	92.06±4.32	92.67±3.95	**95.08±2.12**
11	33.26±7.40	39.06±9.65	89.57±7.29	92.13±4.37	72.49±4.59	71.88±9.16	76.82±7.41	**91.33±2.29**
12	88.50±10.30	95.92±1.22	89.72±4.45	96.43±1.50	50.95±18.59	95.56±3.83	93.08±3.00	**96.38±1.29**
13	82.24±6.11	77.81±0.42	95.52±2.30	94.96±0.27	88.30±1.17	91.33±2.76	92.03±1.33	**94.03±0.82**
14	57.78±27.85	58.35±27.10	100.00±0.00	100.00±0.00	99.23±0.73	99.85±0.34	99.95±0.12	**100.00±0.00**
15	91.06±11.86	95.42±5.36	99.97±0.07	99.90±0.14	99.35±0.36	100.00±0.00	99.87±0.29	**99.74±0.58**
OA(%)	65.74±4.19	64.79±6.35	91.24±0.68	91.73±0.21	84.24±2.08	89.86±1.78	90.69±1.02	**93.54±0.51**
AA(%)	68.74±4.77	66.56±7.66	92.77±0.53	92.49±0.16	86.36±1.77	91.47±1.58	92.09±0.87	**93.90±0.40**
Kappa(%)	62.94±4.54	61.89±6.94	90.52±0.74	91.05±0.22	82.98±2.24	89.03±1.92	89.93±1.10	**93.02±0.55**

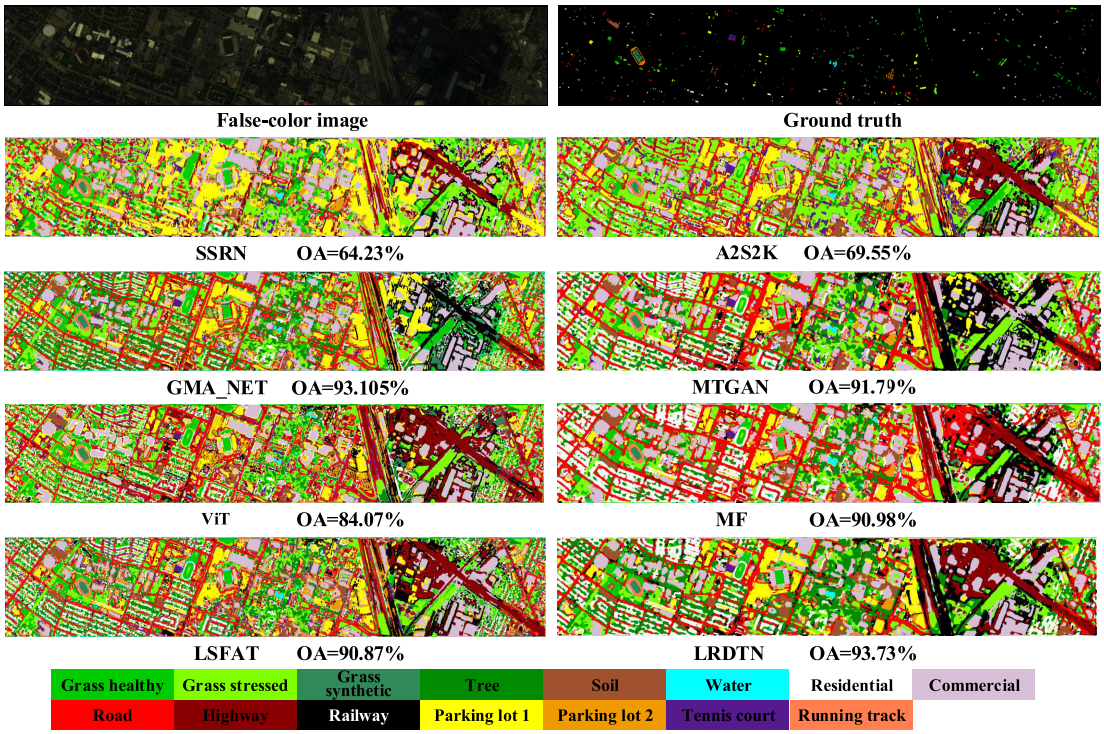


Fig. 9. Classification maps of different methods on the H2013 dataset.

more intuitive and easy-to-interpret visual imagery. This also indicates that when evaluating the performance of the model, we focus on quantitative accuracy metrics and combine the visualization results with the needs of practical application scenarios.

2) Table IV and Fig. 9 Show the Classification Results and Visualized Classification Maps for the H2013 Dataset, Respectively: Table IV shows that the LRDTN model has the highest OA, AA, and Kappa values. Considering the characteristics of the H2013 dataset, which comprises discrete and locally distributed surface objects, where most of the neighboring areas belong to the same class, effectively differentiating this type of local–global spatial relationship is challenging for the model. However, the LRDTN approach demonstrated significant advantages in this case. Specifically, class 8 (commercial) was relatively difficult to classify due

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TABLE V
CLASSIFICATION RESULTS OF THE PU DATASETS

No.	CNN-based method				ViT-based method			
	SSRN	A2S2K	GMA_NET	MTGAN	ViT	MF	LSFAT	LRDTN
1	58.81±3.02	65.71±11.17	92.66±1.69	94.22±4.74	73.30±3.52	87.46±5.21	87.92±4.71	**97.03±1.49**
2	96.89±1.79	97.48±1.13	99.01±0.53	97.85±2.47	85.96±5.73	97.99±1.44	92.38±2.61	**99.40±0.40**
3	70.59±16.09	81.74±6.84	83.29±8.44	79.70±13.55	72.72±5.40	87.74±8.24	72.62±9.19	**94.25±3.90**
4	85.75±2.89	78.99±9.82	98.35±0.72	96.24±0.50	94.55±1.78	94.13±0.96	96.85±0.55	**94.14±2.49**
5	99.18±0.83	99.65±0.14	99.58±0.19	99.88±0.23	99.50±0.46	99.85±0.21	100.00±0.00	**99.92±0.11**
6	49.02±7.63	40.67±3.70	94.34±3.33	93.04±6.75	58.43±13.15	95.15±2.52	61.00±9.59	**97.72±1.83**
7	94.46±1.82	84.89±7.99	94.81±1.96	96.40±2.43	78.95±3.32	98.45±2.06	83.28±8.21	**98.26±2.72**
8	76.42±9.68	78.76±5.36	93.01±3.51	97.27±2.67	75.16±6.52	97.34±1.65	85.52±2.74	**97.10±1.74**
9	97.23±2.36	92.95±4.62	99.96±0.10	99.74±0.24	99.74±0.13	95.55±3.22	99.85±0.06	**97.16±0.44**
OA(%)	81.51±1.03	81.75±2.46	96.06±0.69	95.73±0.83	80.30±1.56	95.21±0.35	86.95±1.06	**97.94±0.24**
AA(%)	80.93±2.13	80.09±3.19	95.00±0.92	94.93±1.04	82.03±1.11	94.85±0.79	86.62±1.76	**97.22±0.63**
Kappa(%)	75.05±1.50	75.03±3.56	94.76±0.93	94.34±1.07	74.14±1.75	93.66±0.45	82.62±1.46	**97.27±0.32**

TABLE VI
CLASSIFICATION RESULTS OF THE KSC DATASETS

No.	CNN-based method			ViT-based method			
	SSRN	A2S2K	MTGAN	ViT	MF	LSFAT	LRDTN
1	96.06±2.91	97.41±1.02	100.00±0.00	96.87±2.08	97.76±2.08	99.49±0.80	**99.89±0.24**
2	60.27±24.98	89.06±6.22	99.19±1.81	73.99±14.31	98.48±1.54	95.96±2.47	**99.82±0.40**
3	68.98±18.06	86.86±10.52	96.95±3.78	72.80±13.25	94.75±5.89	78.64±13.64	**99.92±0.19**
4	36.21±15.76	33.36±17.33	81.64±7.89	29.48±7.78	80.08±12.68	53.88±8.66	**85.26±6.60**
5	49.36±27.26	55.32±23.17	89.79±8.84	42.84±16.15	89.50±14.51	40.14±5.95	**96.74±2.38**
6	62.97±18.86	85.17±25.05	95.22±6.45	60.29±7.66	100.00±0.00	98.81±1.63	**99.90±0.21**
7	97.18±4.45	96.47±3.72	98.12±1.58	88.23±2.35	100.00±0.00	99.76±0.53	**100.00±0.00**
8	66.04±9.19	67.88±10.86	100.00±0.00	92.46±3.90	99.46±0.61	91.77±4.88	**100.00±0.00**
9	87.28±4.29	89.84±5.72	92.20±5.41	77.80±4.88	85.00±6.89	90.88±1.67	**96.24±3.09**
10	49.32±23.61	85.17±10.58	96.19±1.41	95.36±1.72	98.65±1.72	93.85±4.39	**97.40±1.55**
11	93.43±13.85	99.05±1.60	99.75±0.56	99.50±0.59	98.60±1.36	100.00±0.00	**100.00±0.00**
12	66.21±30.54	57.77±15.31	99.17±0.91	90.06±4.42	98.55±1.62	96.81±2.57	**100.00±0.00**
13	94.75±11.73	100.00±0.00	100.00±0.00	100.00±0.00	99.96±0.10	100.00±0.00	**100.00±0.00**
OA(%)	77.19±5.15	84.38±1.54	97.25±0.22	86.02±2.01	96.19±0.53	92.41±1.29	**98.60±0.16**
AA(%)	71.39±5.01	80.27±1.89	96.02±0.66	78.44±2.81	95.44±1.52	87.69±1.88	**98.09±0.37**
Kappa(%)	74.59±5.67	82.55±1.69	96.93±0.25	84.39±2.25	95.74±0.59	91.52±1.44	**98.44±0.17**

to its interpenetration with other surface objects; however, it still achieved the best classification accuracy of 86.94% using the LRDTN method. This result fully verifies the efficiency of the LRDTN in addressing local–global spatial correlation and long-range correlation problems. The visualization results in Fig. 9 show that the proposed method is smoother, with less noise in the classification map and clear classification boundaries, which may be attributed to the fact that the LRDTN model employs DDC and MsEF modules that can consider spatial contextual information, such that the method can better capture the spatial relationships between pixels.

3) Table V and Fig. 10 Show the Classification Results and Visualized Classification Maps for the PU Dataset, Respectively: Table V shows that despite the PU dataset having relatively little spectral band information and the distribution of buildings presenting a high degree of complexity, the LRDTN method successfully overcomes this challenge and achieves excellent classification performance. In contrast, the spectral-spatial residual network (SSRN) method performs most poorly in this respect, especially in the classification of class 6 (bare soil), where it achieves an OA of only 49.02%, which is a significant difference compared to the 97.72% of the LRDTN method. This phenomenon may be due to the limitations of the SSRN method in dealing with complex terrain and mixed-pixel situations, especially in recognizing and distinguishing bare soil and other surface objects with similar spectral characteristics, which makes it difficult to achieve

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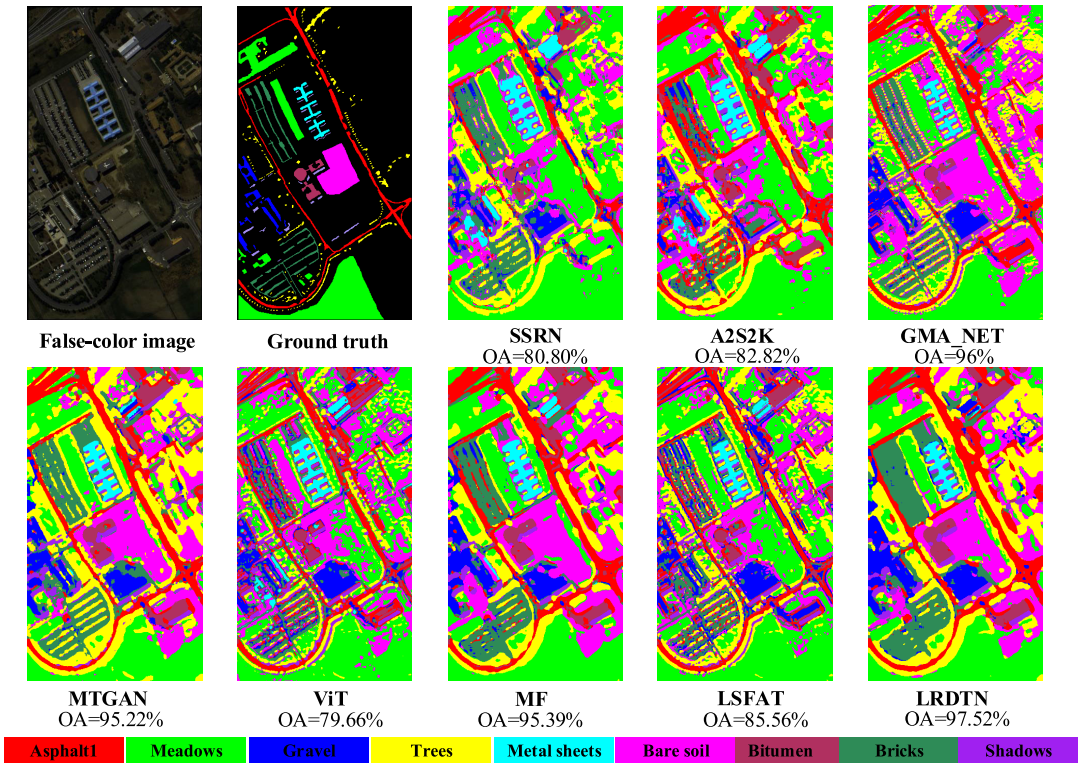


Fig. 10. Classification maps of different methods on the PU dataset.

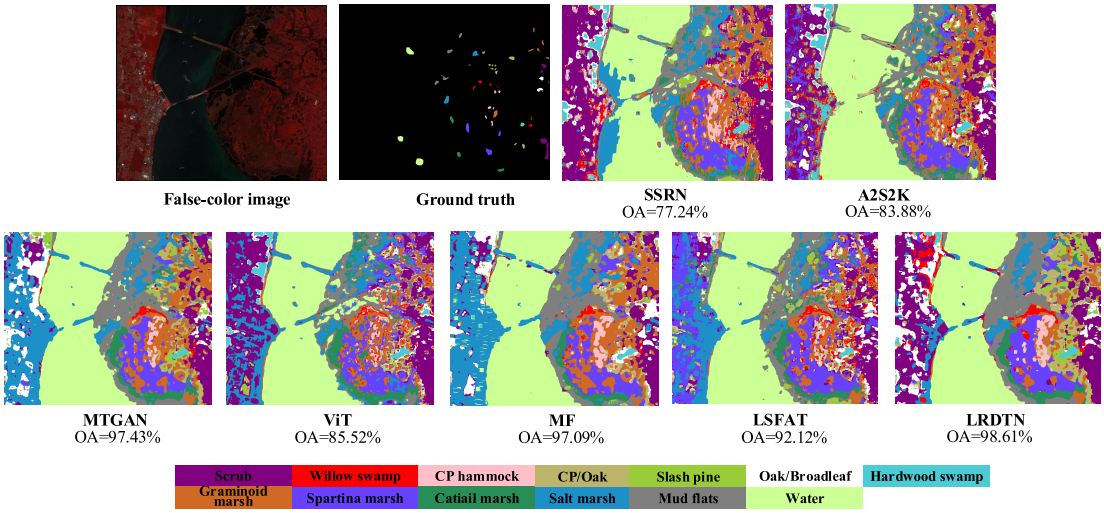


Fig. 11. Classification maps of different methods on the KSC dataset.

high-precision segmentation. The visualized classification map in Fig. 10 shows that the SSRN, A2S2K, ViT, and other methods produce more noise at the classification boundary. However, despite the high overall classification accuracy of the LRDTN method, its edges appear relatively smooth in visual effects. This may be because of the sacrifice of a certain degree of fine-grained information in local regions while optimizing global classification consistency, leading to overly smooth classification boundaries in some scenes.

4) Table VI and Fig. 11 Show the Classification Results and Visualized Classification Maps for the KSC Dataset, Respectively: The KSC dataset contains many marsh areas with highly similar spectral features, which makes classification extremely difficult. According to Table VI, our proposed method shows superior performance in all the evaluation indices. The reason is that the LRDTN successfully integrates the spectral dimension and spatial domain information in the transformer, which enhances the ability of the model to recognize subtle spectral differences in complex scenes. Notably, we designed our experiments using only 20 samples per class for training. Nevertheless, the classification accuracy of the LRDTN on all classes can still maintain a high level of balanced distribution, which strongly validates the effectiveness of the method in dealing with small sample classification tasks. According to the visualized classification map in Fig. 11, existing methods such as SSRN, A2S2K, and LSFAT produce more

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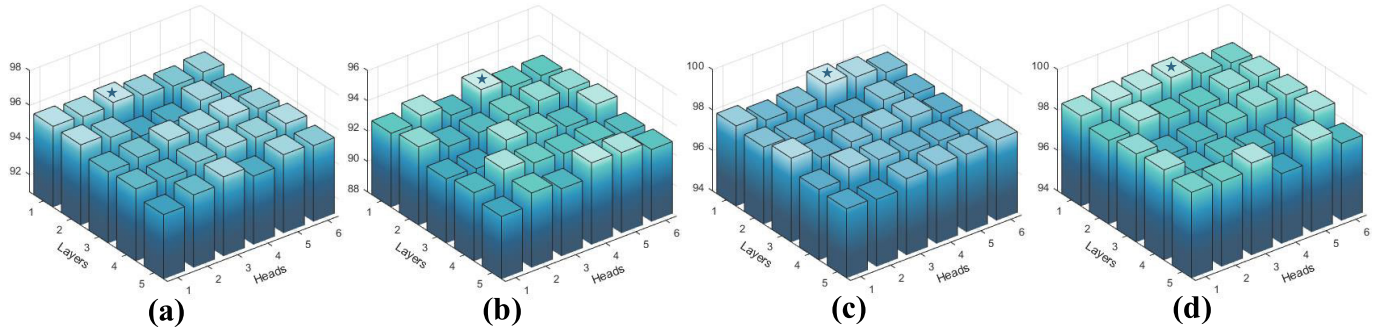


Fig. 12. OA (%) of LRDTN on (a) SV, (b) H2013, (c) PU, and (d) KSC datasets with different numbers of transformer encoder layers and multihead self-attentive heads.

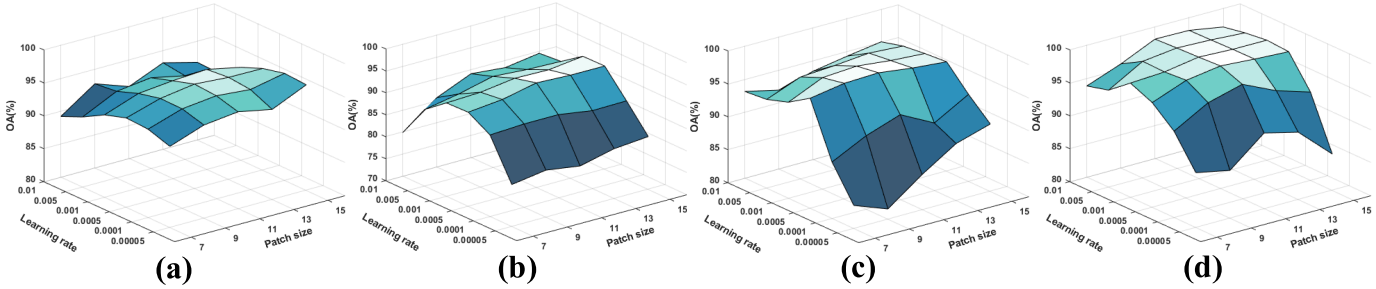


Fig. 13. OA (%) of LRDTN on (a) SV, (b) H2013, (c) PU, and (d) KSC datasets with different learning rates and patch size.

pretzel noise at the classification boundary location. While GMA_NET and MF can suppress such noise to a certain extent and thus obtain more continuous and smooth classification boundaries, this is likely to be at the expense of a certain level of classification accuracy. In contrast, the LRDTN method achieves more desirable classification boundary smoothness under the premise of guaranteeing classification correctness and avoids excessive noise interference.

D. Impact of Different Parameters on the Model

1) Analysis of the Number of Layers and Heads: In most HSI classification methods [40], increasing the number of transformer encoder layers tends to improve the OA of the model, and deeper models are better able to capture the complex relationships and features in the input sequence. However, as shown in Fig. 12, LRDTN can also achieve high OA with only one encoder layer, indicating that it already possesses good representational capabilities on the four datasets, and adding more layers would increase the number of parameters and computational complexity. The model achieves the best OA on three datasets when the number of heads is 4. The best OA is achieved on the SV dataset when the number of heads is 3. Considering that the model cannot adequately capture the long-distance dependence and contextual information in the input sequences when the number of attention heads is too small, we set the number of heads to 4.

In summary, to balance model complexity and performance requirements, we uniformly set the number of layers to 1 and the number of heads to 4.

2) Analysis of Learning Rate and Patch Size: By analyzing Fig. 13(a)–(d) in detail, we note that the OA of the LRDTN increases and then decreases as the learning rate increases from 0.00005 to 0.01 on the four datasets (the learning rate of 5e−4 serves as a threshold). Moreover, as the patch size increases from 7 × 7 to 15 × 15, the classification accuracy of the LRDTN gradually increases, after which it flattens, and finally, it gradually increases.

Based on the above experimental phenomena, we recognize that a larger learning rate may lead to performance degradation while expanding the patch size. However, it can capture more complex structural information and thus improve classification accuracy, increasing the computational and memory overhead. Therefore, our learning rate is set to 5e−4, and the patch size is 11 × 11.

E. Ablation Study

1) Ablation Study of the DDC Module With Different Convolutional Techniques: We designed a series of controlled experiments in which the DDC module was replaced with DwsConv [3] and DyConv [8] to validate the efficacy of the DDC module in HSI classification tasks. In the comparison experiments, we ensured that the convolutional parameter settings (e.g., convolutional kernel size, stride, and padding) used in all models are strictly consistent to ensure fair and comparable results.

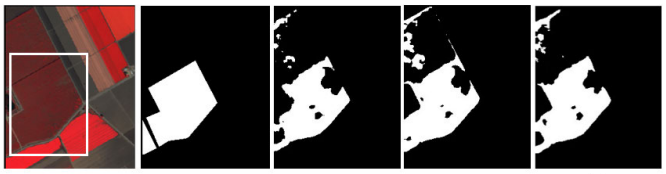
Table VII shows that on the four datasets, DyConv exhibits a higher OA than does DWsConv, especially on the PU dataset, where the OA of DyConv improves by approximately 0.82%. Moreover, DWsConv has a significant advantage over DyConv in terms of the number of training parameters, especially in the SV dataset, where the number of parameters is only one-tenth that of DyConv. The reason for this situation is that DWsConv utilizes a combination of deep convolution and point-by-point convolution, which effectively reduces the parameter size of the model and results in a lightweight design. Finally, our proposed DDC module surpasses the classification accuracy of DyConv in experiments and reduces the number of

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TABLE VII

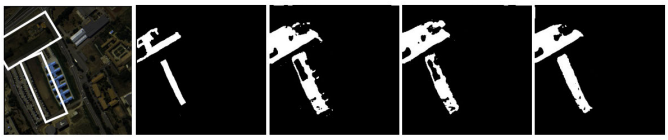
COMPARISON RESULTS OF DIFFERENT CONVOLUTION TECHNIQUES ON DIFFERENT DATASETS. "DW" REPRESENTS THE DEPTHWISE SEPARABLE CONVOLUTION (DWSCONV). "DY" REPRESENTS THE DYNAMIC CONVOLUTION (DYCONV). "DD" REPRESENTS OUR PROPOSED DYNAMIC DEPENDENCE CONVOLUTION MODULE

Methods		Datasets			
		SV	H2013	PU	KSC
Dw	OA(%)	94.32±0.14	91.78±1.04	94.80±1.56	97.41±0.67
	Param(K)	158.85	114.64	89.18	136.57
Dy	OA(%)	94.37±0.27	91.93±0.49	95.62±0.92	97.50±0.31
	Param(K)	1625.21	845.16	462.77	1227.96
DD	OA(%)	95.42±0.08	93.54±0.51	97.94±0.24	98.60±0.16
	Param(K)	224.54	148.05	106.48	185.85



(a) (b) (c) (d) (e)

Fig. 14. Visualization of ablation experiments on different convolutional modules in the SV dataset. (a) False color image. (b) Ground truth. (c) DWsConv. (d) DyConv. (e) DDC.



(a) (b) (c) (d) (e)

Fig. 15. Visualization of ablation experiments on different convolutional modules in the PU dataset. (a) False color image. (b) Ground truth. (c) DWsConv. (d) DyConv. (e) DDC.

TABLE VIII

EXPERIMENTAL RESULTS OF SRA MODULE ABLATION IN MH2SD. "√" REPRESENTS THE PRESENCE OF THE SRA MODULE, AND "×" REPRESENTS THE ABSENCE OF THE SRA MODULE

Methods		Datasets			
		SV	H2013	PU	KSC
√	OA(%)	95.42±0.08	93.54±0.51	97.94±0.24	98.60±0.16
	Param(K)	224.54	148.05	106.48	185.85
	Train(S)	714.67	211.52	429.67	88.41
	Test(S)	3.22	0.75	2.03	0.28
×	OA(%)	95.39±0.21	93.00±0.26	96.97±0.99	97.84±0.36
	Param(K)	227.10	150.61	109.04	188.41
	Train(S)	746.28	216.68	450.45	89.59
	Test(S)	3.35	0.80	2.13	0.29

training parameters compared to DyConv. This phenomenon stems from the unique design of the DDConv module, which skillfully integrates dynamically adjusted weights and efficient convolutional operations and can capture the dynamic dependencies inherent in high-dimensional data while efficiently reducing redundant parameters, improving the performance and efficiency of the model.

In addition, given the relatively discrete distribution of feature categories in the H2013 and KSC datasets, observing subtle differences via direct visualization is easy. Therefore, we chose to visualize part of the SV and PU datasets, where the feature categories are more aggregated. Figs. 14 and 15 show that the classification boundaries are not smooth in the images processed with DWsConv, which reflects that DWsConv fails to integrate local features and contextual information adequately. In contrast, although DyConv improves the smoothness of the classification boundary to a certain extent, it still fails to reach the ideal state in terms of visual effect. In contrast, after adopting our proposed DDC module, the classification results visually present a significant advantage. This proves that the DDC module can flexibly capture the contextual dependencies of the input data and effectively perform local feature modeling in complex visual tasks, improving the overall performance of the model.

2) Ablation Study of the MsEF Module: In the LRDTN, the middle layer of the MsEF is designed with different kernel sizes to capture multiscale features. This multiscale selection concerns the effective integration of local and global information from HSIs, which significantly affects the final classification performance.

According to Fig. 16, on the four datasets, the classification results are relatively poor when the scale sets are {1, 3} and {1, 3, 5}. This is mainly because the smaller scale combinations fail to fully exploit the complementary information of the images at multiple resolutions, resulting in the model's inability to capture the contextual information and global features in the images, which leads to performance degradation. In contrast, when the scale set is changed to {1, 3, 5, 7, 9}, the classification accuracy gradually decreases, especially on the H2013, PU, and KSC datasets, which is more obvious. This is because a larger scale range introduces redundant information. In addition, a larger scale range increases the computational overhead and the number of parameters in the model. Therefore, to make a tradeoff between performance and computational efficiency, we chose the scale set of {1, 3, 5, 7} for training to achieve better classification performance.

3) Ablation Study of the SRA Module in MH2SD: In Section III-C1, we propose the SRA mechanism introduced by the MH2SD method [43]. This mechanism efficiently refines the input features through a spatial dimension compression strategy to eliminate unnecessary redundant information and reduce the network parameters before entering the 2SD processing stage. We conducted ablation experiments and evaluated several aspects, including the OA, number of parameters, training time, and testing time to evaluate the impact of the SRA module on the overall performance. The results of the experiments are detailed in Table VIII.

We observe that the method incorporating the SRA module exhibits a significant improvement in classification accuracy. This indicates that the SRA module can effectively extract critical features and reduce the interference of noncritical features in the classification task. Additionally, we note that the method incorporating the SRA module achieves effective streamlining in terms of the number of network parameters. This further validates the feature compression function of the

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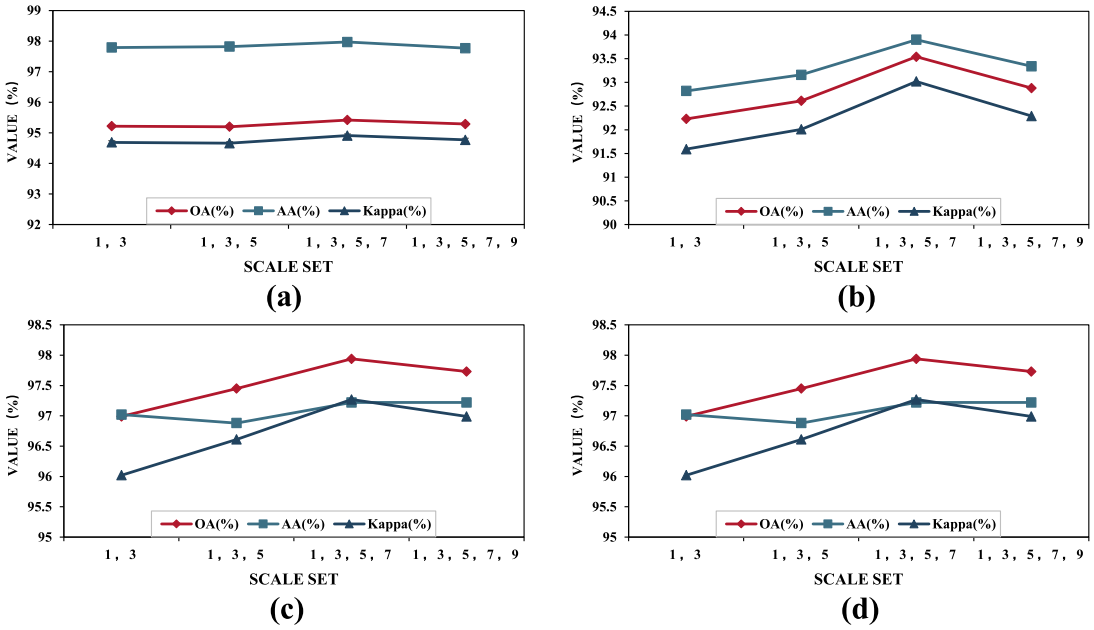


Fig. 16. Ablation study of MsEF module at different scales on four datasets. (a) SV. (b) H2013. (c) PU. (d) KSC.

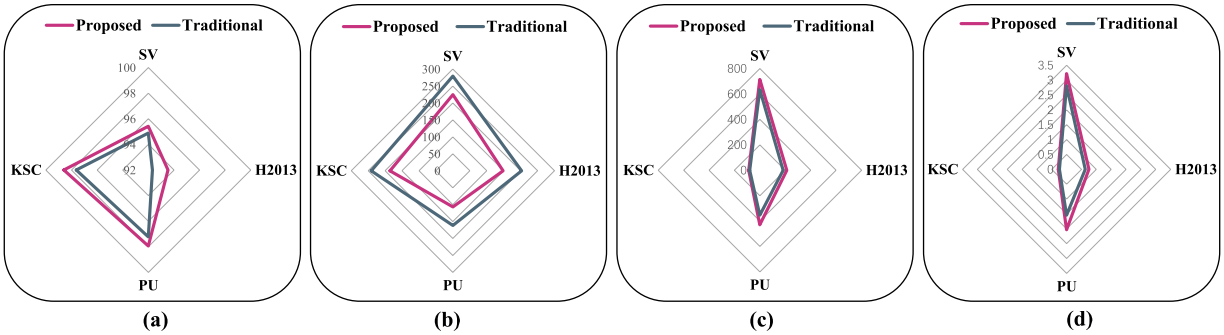


Fig. 17. Radar plots of OA, network parameters, training time, and testing time of the proposed LGPT with conventional ViT under four datasets SV, H2013, PU, and KSC. (a) OA (%). (b) Network parameters (K). (c) Training time (s). (d) Testing time (s).

SRA module, which reduces the complexity of the model by removing redundant information and reducing the computational cost of model training and inference.

4) Ablation Study of the LGPT Module: We compare it with ablation experiments conducted by the traditional ViT [13] to demonstrate the usefulness and improvement of the LGPT proposed in this article. The LGPT proposed in this article is shown in Fig. 5, while the traditional multihead attention mechanism is shown in Fig. 6(a). The parameters involved in both methods (e.g., the number of attention mechanism heads, the number of layers, etc.) are kept consistent during the experiments.

First, we observe Fig. 17(a). The proposed method outperforms the traditional method in terms of classification accuracy, as the OA of the proposed method is 93.54% on the H2013 dataset. The OA of the traditional method is 92.32%. This indicates that the proposed method is an effective method to compensate for the lack of local modeling by introducing convolutional operations and spectral information, which makes the model better able to extract local–global information and thus improves the accuracy of the HSI processing task.

Second, Fig. 17(b) shows that the number of network parameters in the traditional method is much greater than that of the proposed method. Taking the H2013 dataset as an example, the network parameters of the traditional method are 203.06 K, while those of the proposed method are 148.05 K, which is a reduction of approximately 55 000 parameters. This validates the performance of the FCE in the LGPT proposed in this article (in Section III-C2), and by replacing the MLP in the traditional ViT, the FCE further realizes the deep fusion of the CNN and ViT and reduces the number of network parameters.

Finally, Fig. 17(c) and (d) shows that the two curves almost overlap, indicating that the gap between the two methods in terms of time consumption is quite limited. This moderate increase in time cost is acceptable considering the accuracy improvement and the parameter reduction advantage of the new method.

5) Ablation Study of the Proposed Module: To validate the effectiveness of the components of the LRDTN algorithm, we conducted a series of ablation experiments on DDC, MsEF, and LGPT. In Table IX, a check mark ($\surd$) indicates that the component is included in the experiment.

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TABLE IX
CLASSIFICATION PERFORMANCE IN ABLATION EXPERIMENTS (THE BOLD TEXT REPRESENTS THE RESULTS OF THE PROPOSED METHOD)

Datasets	DDC	MsEF	LGPT	OA	AA	Kappa
SV	√	√	√	**95.42±0.08**	**97.97±0.16**	**94.91±0.09**
	√	√		94.95±0.04	97.58±0.22	94.38±0.05
	√		√	94.99±0.14	97.60±0.14	94.42±0.16
		√	√	95.01±0.30	97.61±0.49	94.45±0.34
H2013	√	√	√	**93.54±0.51**	**93.90±0.40**	**93.02±0.55**
	√	√		92.16±0.30	92.80±0.28	91.52±0.32
	√		√	91.39±0.67	92.09±0.52	90.69±0.73
		√	√	91.95±0.87	92.64±0.70	91.29±0.94
PU	√	√	√	**97.94±0.24**	**97.22±0.63**	**97.27±0.32**
	√	√		96.72±0.77	96.18±0.59	95.66±1.00
	√		√	97.01±0.63	96.00±1.34	96.03±0.83
		√	√	95.45±2.47	96.49±0.69	94.05±3.18
KSC	√	√	√	**98.60±0.16**	**98.09±0.37**	**98.44±0.17**
	√	√		97.67±0.33	96.72±0.55	97.40±0.36
	√		√	97.31±0.61	96.51±0.69	96.99±0.68
		√	√	97.23±0.82	96.00±1.23	96.91±0.91

It can be observed that the classification accuracy is relatively low without the emphasis DDC module. In contrast, the classification accuracy further improves on the PU and KSC datasets when the DDC module is added (without MsEF). This indicates that the DDC module effectively captures the contextual features of the input data through dynamic dependence modeling. When the DDC and MsEF are added (LGPT module is not used), the accuracy is still noticeably higher, as the MsEF module refines and integrates the dynamic features extracted by the DDC module at multiple levels. Hence, the MsEF module enhances the overall expressive capability.

Finally, when the LGPT module is included (DDC + MsEF + LGPT), there is a significant improvement in OA, AA, and Kappa metrics across the four datasets. This demonstrates that the three components—DDC, MsEF, and LGPT—collectively enhance classification performance and are indispensable.

F. Model Complexity Analysis

In Table X, the LRDTN obtains better classification results with the least number of network parameters. On the PU dataset, the LRDTN has only 0.11M parameters. Specifically, the training time of our proposed method is only second to MTGAN primarily because the LRDTN network structure is relatively complex and includes two branches, which requires more time for training compared to MTGAN. Additionally, the Flops of LRDTN are slightly higher than those of LSFAT yet its training time is lower. This may be attributed to the efficient architectural designs employed in LRDTN, such as the DDC and MsEF modules, which maintain high computational efficiency while reducing the number of parameters. Although the training time of the LRDTN on the four datasets is not optimal, the LRDTN achieves higher classification accuracy than the other algorithms, as shown in Tables III–VI. Taken together, the LRDTN method achieves better efficiency through parameter optimization while maintaining classification accuracy.

TABLE X
MODEL COMPLEXITY OF THE PROPOSED LRDTN AND COMPARISON METHOD (THE BOLD TEXT REPRESENTS THE RESULTS OF THE PROPOSED METHOD)

Datasets	Metric	SSRN	A2S2K	GMA-NET	MTGAN	ViT	MF	LSFAT	LRDTN
SV	Train(S)	2261	2778	1020	528	1030	2195	810	**712**
	Test(S)	9.7	12.1	4.7	2.3	4.7	10.6	3.7	**3.2**
	Flops(G)	19.34	20.95	5.00	1.66	1.60	2.91	0.39	**0.86**
	Param(M)	0.37	0.38	1.79	0.34	0.36	0.28	0.28	**0.22**
H2013	Train(S)	575	677	228	156	242	556	249	**212**
	Test(S)	1.8	2.3	0.8	0.5	0.9	2.4	0.9	**0.8**
	Flops(G)	16.97	18.38	3.39	1.52	1.41	2.56	0.47	**0.78**
	Param(M)	0.28	0.28	1.5	0.38	0.23	0.23	0.28	**0.15**
PU	Train(S)	865	1050	468	276	470	1290	500	**430**
	Test(S)	3.9	4.9	2.2	1.2	2.2	6.3	2.4	**2.0**
	Flops(G)	8.52	9.22	1.56	0.80	0.71	1.51	0.32	**0.42**
	Param(M)	0.22	0.22	1.37	0.34	0.17	0.23	0.28	**0.11**
KSC	Train(S)	285	327	-	68	106	216	103	**88**
	Test(S)	0.7	1.6	-	0.2	0.4	0.9	0.4	**0.3**
	Flops(G)	12.49	13.53	-	1.09	1.03	1.88	0.29	**0.56**
	Param(M)	0.33	0.33	-	0.41	0.30	0.26	0.28	**0.19**

G. T-SNE Visualization and Analysis

To further validate the effectiveness of the LRDTN method, the t-distributed stochastic neighbor embedding (T-SNE) technique is employed on four datasets to visualize the feature distribution of various methods. Additionally, referring to [57], the Fisher discriminant analysis method is used to calculate the interclass scatter coefficient S_b, with the corresponding calculation results displayed below the feature distribution plots. The experimental results are shown in Figs. 18–21.

In SV (Fig. 18) and H2013 (Fig. 19) datasets, ViT and LSFAT exhibit severe label mixing, while SSRN, A2S2K, GMA_NET, and MTGAN demonstrate a certain degree of intraclass scatter, especially MTGAN, which shows a distinct banding pattern with larger intraclass distances. In contrast, our method exhibits less label mixing and visually demonstrates better intraclass and interclass dispersion. In the PU dataset (Fig. 20), the feature distributions of SSRN, A2S2K, GMA_NET, and MTGAN are distinctly band-like, with a large intraclass distance. The ViT method performs poorly, severely mixing classes 1, 3, 7, and 8. In contrast, the LRDTN method achieves effective interclass clustering and significantly reduces label mixing, while maintaining ideal interclass distances. In the KSC datatset (Fig. 21), all methods except for ViT have a large interclass distance. However, compared to these control methods, the LRDTN method is superior in terms of intraclass clustering.

When using quantitative methods to measure the interclass dispersion of different models, it is observed that the SSRN and A2S2K models exhibited the highest interclass dispersion S_b across four datasets, followed by the LRDTN model. This is because during T-SNE visualization, the SSRN and A2S2K models mostly show a banded distribution or intraclass dispersion. When calculating S_b, the distance between the centers of each class are primarily considered. This banded distribution indicates that the data have a larger expansibility in certain directions, leading to an increase in intraclass distance. Nevertheless, the LRDTN model performed equally well, or possibly better, in maintaining interclass separation. This is because the LRDTN model achieves a tighter intraclass

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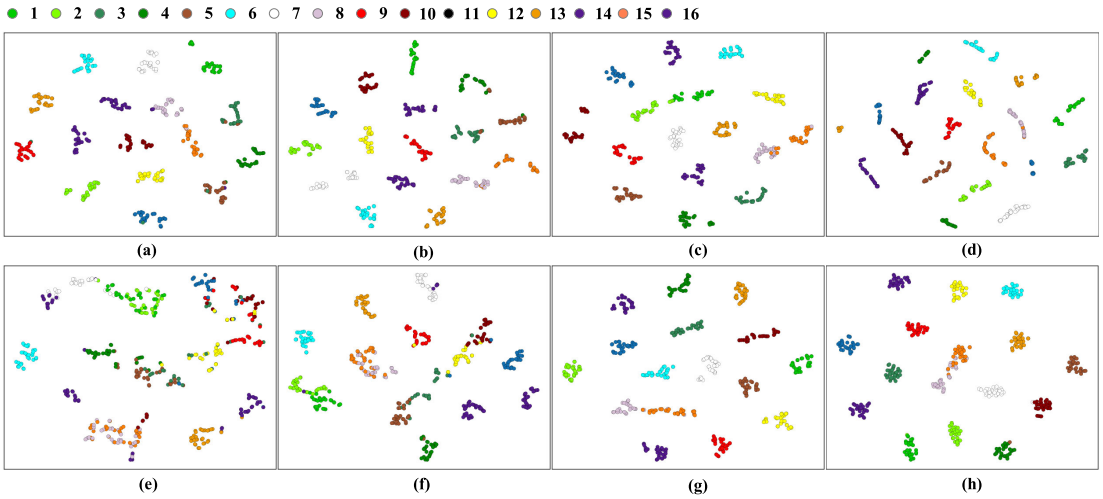


Fig. 18. Visualization of T-SNE using different methods on the SV dataset. (a) SSRN ($S_b = 1.09$). (b) A2S2K ($S_b = 1.14$). (c) GMA-NET ($S_b = 1.12$). (d) MTGAN ($S_b = 0.75$). (e) ViT ($S_b = 0.48$). (f) LSFAT ($S_b = 0.66$). (g) MF ($S_b = 1.06$). (h) LRDTN ($S_b = 1.12$).

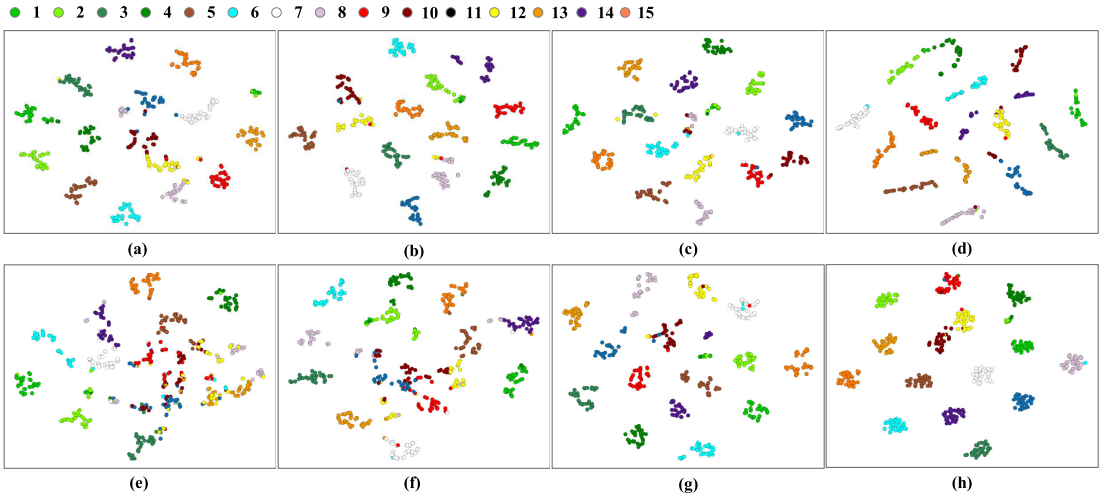


Fig. 19. Visualization of T-SNE using different methods on the H2013 dataset. (a) SSRN ($S_b = 1.04$). (b) A2S2K ($S_b = 1.03$). (c) GMA-NET ($S_b = 1.04$). (d) MTGAN ($S_b = 0.72$). (e) ViT ($S_b = 0.65$). (f) LSFAT ($S_b = 0.75$). (g) MF ($S_b = 0.93$). (h) LRDTN ($S_b = 1.02$).

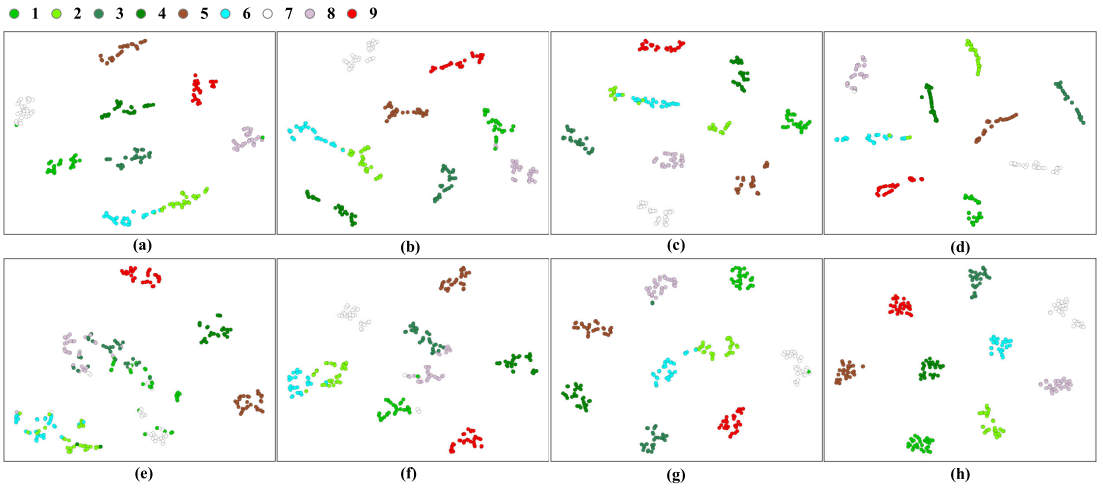


Fig. 20. Visualization of T-SNE using different methods on the PU dataset. (a) SSRN ($S_b = 1.18$). (b) A2S2K ($S_b = 1.14$). (c) GMA-NET ($S_b = 1.14$). (d) MTGAN ($S_b = 0.89$). (e) ViT ($S_b = 0.85$). (f) LSFAT ($S_b = 0.96$). (g) MF ($S_b = 1.09$). (h) LRDTN ($S_b = 1.13$).

clustering, which helps to clearly define the boundaries of each classification.

In summary, LRDTN can achieve the maximization of inter-class distance and the minimization of intraclass distance in the

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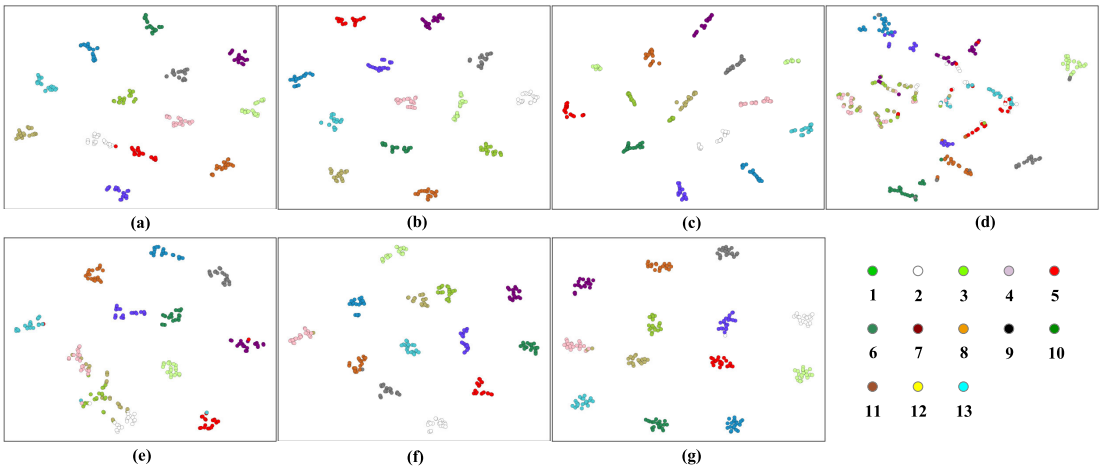


Fig. 21. Visualization of T-SNE using different methods on the KSC dataset. (a) SSRN ($S_b = 1.2$). (b) A2S2K ($S_b = 1.23$). (c) MTGAN ($S_b = 0.98$). (d) ViT ($S_b = 0.56$). (e) LSFAT ($S_b = 0.71$). (f) MF ($S_b = 1.05$). (g) LRDTN ($S_b = 1.07$).

construction of feature space. This demonstrates that LRDTN can effectively describe the long-range global dependencies among HSI samples, playing a significant role in improving the classification performance of HSI.

V. Conclusion

A neural network architecture called the LRDTN is proposed. The DDC and MsEF modules are mainly utilized to efficiently capture and integrate the contextual relationships within HSI features and simultaneously process spectral–spatial information at different scales. Moreover, a novel transformer variant, LGPT, is designed to enhance the ability of the model to understand and utilize HSI local features and global contextual features and their complex long-range dependencies. Inside this structure, MH2SD enables the model to construct local feature representations from a global perspective accurately, and the FCE module is used to replace the traditional MLP, drastically reducing the number of network parameters while deepening the fusion of local and global features. We conducted comprehensive experimental validation of the LRDTN on four hyperspectral datasets. The experimental results show that the overall performance of our method outperforms that of several comparative methods in all cases and can still significantly outperform the classification performance of the other models, even when the number of parameters is limited.

Future research directions will explore ways to further integrate graph convolutional networks (GCNs). Although combining CNNs with transformers enables the model to effectively capture local–global information about HSI and model long-term dependencies more efficiently, it is still less effective when dealing with small quantities of sample data. Our research has found that GCNs exhibit advantages in handling small sample data, particularly in capturing local structures; however, their ability to represent global information is limited. Additionally, the construction of graph networks introduces high computational costs. To address these issues, we will explore lightweight HSI processing methods that integrate GCNs, specifically optimized for small sample data, with the aim of further improving the performance of HSI classification.

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